of the varactors varied from 1.8 to 16 pF. The resonant frequency shifted from 160 to 200 MHz. The VSWR was kept less than 1.3 from 188 to 192 MHz. It was confirmed that the prototype antenna on glass covered the digital terrestrial radio band.

4. CONCLUSIONS

A varactor-loaded compact folded dipole antenna has been proposed for digital terrestrial radio reception. The prototype antenna in free space showed a VSWR of less than 1.4 and gain between −1.4 and −3.1 dBi from 187 to 230 MHz. The VSWR remained less than 1.3 in the digital terrestrial radio band from 188 to 192 MHz, even when the antenna was attached to the glass. The proposed antenna has advantages of a compact structure of 332 mm by 30 mm, wide tuning range with low VSWR, and ease of fabrication. It will be used for digital terrestrial radio reception in vehicles.

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THEORETICAL AND EXPERIMENTAL STUDY OF POLARIZATION CHARACTERISTICS OF POLARIZATION MAINTAINING FIBER BASED ON WAVELENGTH-SWEEPING MODULATION

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ABSTRACT: *We analyze theoretically the polarization characteristics of polarization maintaining fiber and study the basic measurement principles of beat length and polarization extinction ratio of this kind of optical fiber. According to the dependence of the phase difference between two orthogonally polarized modes (denoted as HE^x_{11} and HE^y_{11}) transmitted in the polarization maintaining fiber on the light wavelength, we propose the wavelength-sweeping modulation method to measure the beat length and the model birefringence. Based on this technique, the beat length and polarization extinction ratio of the PANDA polarization maintaining fibers (PMFs) (provided by Yangtze Optical Fiber and Cable Company, Wuhan, China) were investigated in detail. Experimental results show good consistent with the theoretical ones. We find that this method shows high measurement precision with the advantages of clear measurement principle and easy to operate.* © 2010 Wiley Periodicals, Inc. Microwave Opt Technol Lett 52: 1466–1469, 2010; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.25244

Key words: *polarization maintaining fiber; birefringence; beat length; polarization extinction ratio*

1. INTRODUCTION

Polarization maintaining fibers (PMFs) are expected to play an important role for instance in fiber lasers, fiber sensors, and fiber gyroscopes. These optical devices require that the polarization state of the received signal must be kept constant. To meet this requirement, various PMFs have been developed, for example PANDA [1, 2], bow-tie [3] (where the birefringence is due to stress in the core region), and polarization maintaining photonic crystal fiber [4] (the birefringence is due to geometrical form).

Conventional PMF is characterized by its model birefringence and polarization extinction ratio. The degree of model birefringence is often expressed by the beat length between two orthogonally polarized modes. Lots of methods have been proposed to measure the beat length L_B of PMF. Among those proposed methods, the cutback method [5] is destructive; the twist method [6], the electro- or magneto-optical modulation method [7], the lateral force method [8] become difficult when the beat length L_B is less than 10 mm; the Rayleigh scattering method [9] and the optical frequency domain reflectometer method [10] are also difficult to perform when the fiber loss is small. And the most widely used method for beat length measurement is the scheme with the combination of the crossed polarizer technique [11] and the wavelength scanning method [12]. The principle of this scheme is mainly based on the dependence of the phase difference between the two orthogonally polarized modes in the fiber on the wavelength of the light. As usual, broadband Amplified Spontaneous Emission (ASE) is used in this scheme which results in a relative higher measurement error of the beat length due to the group dispersion in the fiber.

In this article, we theoretically analyze and experimentally study the polarization properties of the PMFs by adopting the similar measurement scheme but using the high quality tunable laser source instead of ASE. The beat length and the mode birefringence of the PANDA PMFs were measured accurately. The experimental polarization extinction ratio shows good consistency with the theoretical result. It can be seen that this so-called wavelength-sweeping modulation method shows high measurement precision with the advantages of clear measurement principle and easy to operate.

2. PRINCIPLE

In general, the polarization property of the PMF is described by its model birefringence, while the model birefringence is obtained by measuring the beat length between two orthogonally polarized modes (denoted as HE_{11}^{x} and HE_{11}^{y}). The relationship between the polarization beat length L_B and the model birefringence B is given by Ref. 13.

$$L_B = \frac{\lambda}{n_x - n_y} = \frac{\lambda}{B} = \frac{2\pi}{\beta_x - \beta_y} \tag{1}$$

where λ is the wavelength of the light in vacuum, $n_x(\beta_x)$ and $n_y(\beta_y)$ are the effective refractive index (propagation constant) for the two polarization modes polarized along x- and y-axis, respectively. It indicates that shorter L_B corresponds to stronger model birefringence.

Suppose a linear polarization light is launched upon the PMF's input end, the phase difference between the two modes of HE_{11}^{x} and HE_{11}^{y} at the output end can be expressed as

$$\phi = \frac{2\pi}{\lambda}(n_x - n_y)l = \Delta\beta l \tag{2}$$

where $\Delta\beta$ denotes the difference in propagation constants of the two modes, l is the fiber length.

If the wavelength λ of the light source is swept by a small amount $\Delta\lambda$, where $\Delta\lambda \ll \lambda$, then from Eq. (2) the change of the phase difference ϕ induced by $\Delta\lambda$ is given by

$$\Delta\phi = -\frac{2\pi(n_x - n_y)l\Delta\lambda}{\lambda^2} = -\frac{2\pi l \Delta\lambda}{L_B \lambda}. \tag{3}$$

In Eq. (3), it is assumed that $\Delta\beta \propto \lambda^{-1}$, the effects of material and waveguide dispersions have been neglected [14].

To measure the dependence of the phase difference between two orthogonally polarized modes of HE_{11}^{x} and HE_{11}^{y} on the wavelength of the injected laser, as shown in Figure 1, we select the two orthogonal principal axes of the PMF to be parallel to the coordinate axes. The orientations of the linear polarization light and the Wollaston prism's optical axes are tilted θ and β with respect to the principal axis x of the PMF, respectively. Light intensities of the two output beams, respectively, polarized along the Wollaston prism's optical axes (C_1 and C_2), via the Wollaston prism are given as

$$\begin{cases} I_{C_1} = I_0^2(\cos^2\theta\cos^2\beta + \sin^2\theta\sin^2\beta \\ \qquad\qquad + 2\cos\theta\cos\beta\sin\theta\sin\beta\cos\phi) \\ I_{C_2} = I_0^2(\cos^2\theta\sin^2\beta + \sin^2\theta\cos^2\beta \\ \qquad\qquad - 2\cos\theta\cos\beta\sin\theta\sin\beta\cos\phi). \end{cases} \tag{4}$$

For simplicity, we choose $\theta = \beta = 45°$ to make Eq. (4) take the simple form as follows:

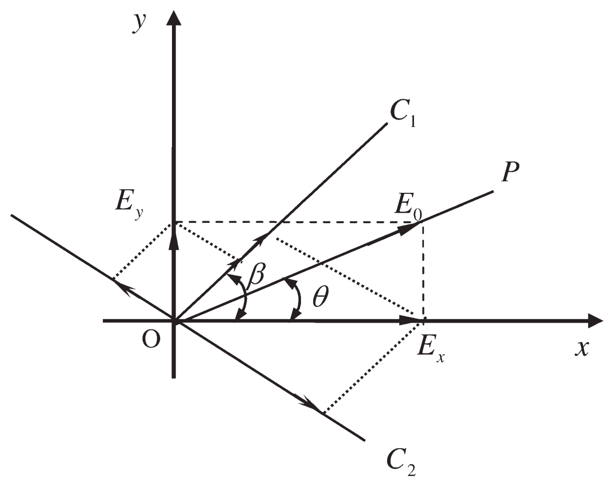

Figure 1 Schematic diagram of polarization decomposition. C_1 and C_2 are the two optical axes of Wollaston prism. P indicates the polarization orientation of the linear polarization light, which has an angle θ with respect to the principal axis x of the polarization maintaining fiber. E_0 is the electric intensity of the linear polarization light. β is the angle between one of the optical axes of Wollaston prism and the principal axis x of the polarization maintaining fiber

$$\begin{cases} I_{C_1} = I_0^2(1 + \cos\phi)/2 \\ I_{C_2} = I_0^2(1 - \cos\phi)/2 \end{cases}. \tag{5}$$

Thus, from Eq. (5) we have

$$\sin\phi = \pm\frac{2\sqrt{I_{C_1}I_{C_2}}}{I_{C_1} + I_{C_2}}. \tag{6}$$

So the phase difference ϕ between two orthogonally polarized modes at the output end of the PMF from Eq. (6) can be written as

$$\phi = n\pi \pm \arcsin\left(\frac{2\sqrt{I_{C_1}I_{C_2}}}{I_{C_1} + I_{C_2}}\right) \tag{7}$$

where n is an integer. Thus, by measuring the quantity of the intensities (I_{C_1} and I_{C_2}) as a function of the phase difference, we can get the $\phi \sim \lambda$ relation, the beat length and the model birefringence can be obtained by using Eq. (3).

Polarization extinction ratio is another important factor of PMF. If we launch the linear polarization light into the PMF with its polarization orientation parallel to one of the principal axes of the fiber, as usual, due to polarization crosstalk there will excite an orthogonally polarized mode with its polarization orientation parallel to another principal axis of the fiber. Suppose the power of the two orthogonally polarized modes at the back of the Wollaston prism are p_1 and p_2, respectively, then the polarization extinction ratio can be defined as

$$\eta = 10\left|\log\frac{p_2}{p_1}\right|(dB). \tag{8}$$

In theory, when the Wollaston prism is arranged with its optical axes to be parallel to the principal axes of the PMF, thus we have $\sin\beta = 0$ or $\cos\beta = 0$. According to Eq. (4), the intensities of the two orthogonally polarized modes become

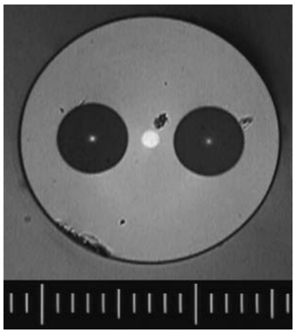

Figure 2 Micrograph cross section of PANDA fiber. The smallest scale is 10 μm

$I_0^2 \cos^2 \theta$ and $I_0^2 \sin^2 \theta$, respectively. From Eq. (8), we obtain the theoretical polarization extinction ratio.

$$\eta = 10\left|\log(ctg^2\theta)\right|(dB). \tag{9}$$

3. EXPERIMENTAL RESULTS

The PMF samples of PANDA with stress birefringence are provided by Yangtze Optical Fiber and Cable Company, Wuhan, China, and its standard beat length is 3.5 mm at 1550 nm. Figure 2 is the micrograph cross section of the PANDA fiber. The measurement experimental setup is shown schematically in Figure 3. The principal axes at both ends of the PMF fiber are firstly determined, as shown in Figure 2, the horizontal orientation is parallel to the horizontal principal axis of the PMF. A tunable laser provided by EXFO Company is used as optical source, the wavelength of light can be continuously tuned from 1500 to 1650 nm with very high precision of 0.001 nm. The amplified laser by EDFA is launched into PMF through Polarization Beam Splitter (PBS). By rotating the PBS, linear polarization light with different polarization orientation can be obtained. When polarized light is incident at 45° to the principle axes of PMF, the output light polarization state can be analyzed with a Wollaston prism also at 45° to the PMF principal axes. The wavelength is tuned from 1547 to 1553 nm with a step of 0.1 nm, the output light powers (p_1 and p_2) of the orthogonally polarized modes are measured by two power meters having equal sensitivity. The variations of two output beams with wavelength are shown in Figure 4. This experimental result is agreement with the theoretical prediction from Eq. (5). The variation of the phase difference between two orthogonal polarization states with wavelength is shown in Figure 5. By linearly fitting the dependence of $\Delta\phi$ on $\Delta\lambda$, we can easily calculate the beat length of the PMF from Eq. (3). Thus the beat length is 3.56 mm at the 1550 nm, and the corresponding birefringence is 4.37×10^{-4}.

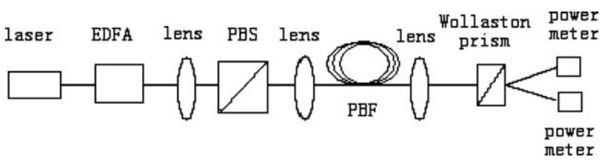

Figure 3 Schematic diagram of the experimental setup. EXFO is the tunable laser source with the wavelength range from 1500 to 1650 nm and with the tunable precision of 0.001 nm erbium doped fiber amplifier (EDFA), polarization beam splitter (PBS), polarization maintaining fiber (PMF)

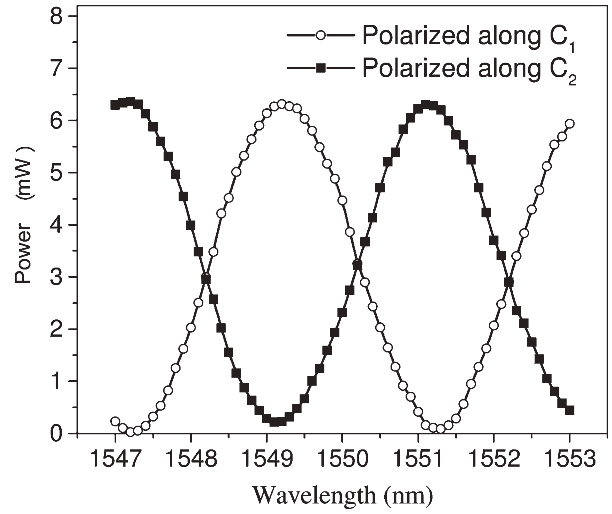

Figure 4 Power variations of two output beams via Wollaston prism with laser wavelength. Here the two light beams polarize along the two optical axes C_1 and C_2 of the Wollaston prism, respectively

The polarization extinction ratio can also be measured by adopting the same experimental setup in Figure 3. Here, the optical source is operated at wavelength of 1550 nm and the Wollaston prism is tilted 0° with respect to the principal axes of PMF. When the linear polarization light is launched at different tilted angle of θ to the principal axes of PMF by rotating the PBS, the output light power (p_1 and p_2) of the two orthogonally polarized modes can be detected at the back of Wollaston prism. The dependence of the output polarization extinction ratio on the input polarization angle is shown in Figure 6 according to Eq. (8). The solid line in Figure 6 is the result of theoretical calculation. This shows that there is good agreement between the calculated and experimental results and the maximum extinction ratio in accordance with the fast- or slow-axis of the fibers means the cross-talk. From the results of the extinction ratio measurement, the sample fiber had the largest cross-talk of −32.8 dB.

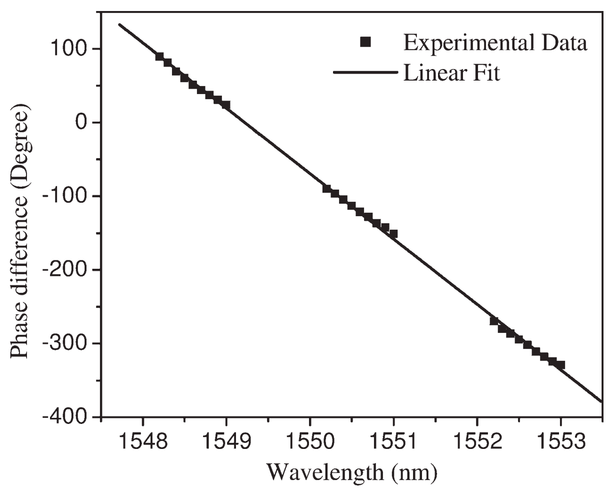

Figure 5 Variation of the phase difference between the two orthogonal polarized states via the polarization maintaining fiber with wavelength. The linear fit is consistent with the dependence of $\Delta\phi$ on $\Delta\lambda$ described by Eq. (3)

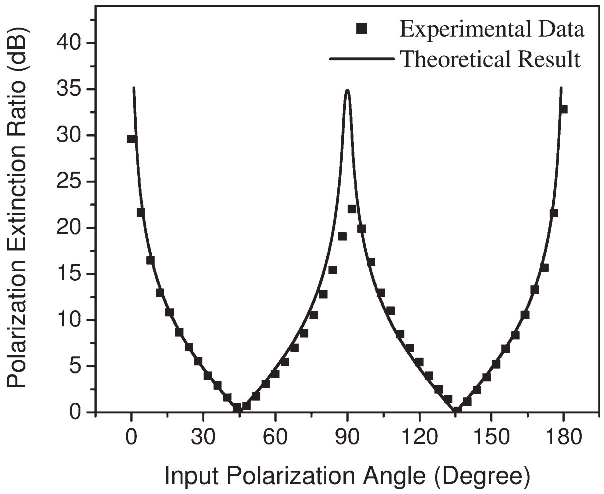

Figure 6 Dependence of the polarization extinction ratio of the two orthogonal polarized beams on the polarized orientation angle of the input linear polarized light

4. CONCLUSIONS

We analyze the polarization properties of polarization maintaining fiber and study the basic measurement principles of beat length and polarization extinction ratio. Adopting the high accurate tunable laser source and using the wavelength-sweeping modulation method, the beat length and polarization extinction ratio of PANDA fiber is measured. The experimental results indicate, at the working wavelength of 1550 nm, the largest cross-talk of the sample fiber is -32.8 dB, the beat length is 3.56 mm and the corresponding model birefringence is 4.37×10^{-4}, and there is good agreement between the calculated and experimental results.

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LOW-COST SURFACE-REDUCTION TECHNIQUE FOR RFID READER ANTENNAS

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ABSTRACT: *A low-cost technique for surface reduction of patch antennas is presented. It is based on the slow wave phenomenon. To create slow wave, a frequency selective surface has been burned on the ground plane of a patch antenna. With this technique, a patch antenna has been designed. An antenna for RFID reader at 915 MHz has been designed and simulated. It uses a 1.6-mm-thick FR4 epoxy substrate ($\varepsilon_r = 4.5$ h $= 1.6$ mm tan (δ) $= 0.02$) measures 61 mm × 65 mm. His maximum gain is 2 dB.* © 2010 Wiley Periodicals, Inc. Microwave Opt Technol Lett 52: 1469–1471, 2010; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.25241

Key words: *RFID; antenna; frequency selective surfaces; slow wave*

1. INTRODUCTION

The grow of applications based on contactless radio identification in the UHF frequency band demands new research efforts for antenna designers. From this perspective, two of the main antenna design challenges are the reduction of the cost production and their miniaturization. For tag antennas, many low cost solutions of miniaturization have already been proposed [1]. Conversely, for 50 Ω UHF RFID reader antennas, few solutions for miniaturization have been published. In some RFID applications, a portable reader will be helpful. In the most important part of commercial readers, patch antennas are used, but this topology in the UHF frequency band suffers of large area which remains portability difficult. This involves reduction of the antenna size without production cost augmentation to achieve portability of UHF RFID readers. Increases the effective dielectric permittivity $\varepsilon_{\rm eff}$ allows the miniaturization of antennas. Several solutions can be investigated, for example use of ceramics with high ε_r but it suffers of high cost. Some authors propose to build specific substrates based